

Cultural Specific Item Recogniser

Natural Language Processing
IIT Hyderabad

Problem Statement

Problem

Standard NER models fail to detect **culture-specific entities**

- Examples: food, rituals, festivals, clothing
- Indian cultural data is underrepresented

Goal

Build a **Cultural Specific Item Recogniser**

Approach

- Dataset: DIWALI (HuggingFace)
- Template-based data augmentation
- BIO tagging for sequence labeling
- Fine-tuned BERT for token classification
- Train/Validation/Test split used

Pipeline

Data → Augmentation → Tokenization → Training → Evaluation

Dataset

- Original dataset: 8,800 samples
- Generated dataset: **12,000 samples**
- Augmented using templates for diversity
- Split:
 - Train: 9600
 - Validation: 1200
 - Test: 1200
- Annotation: BIO format

Example

Gondi is part of Indian tradition

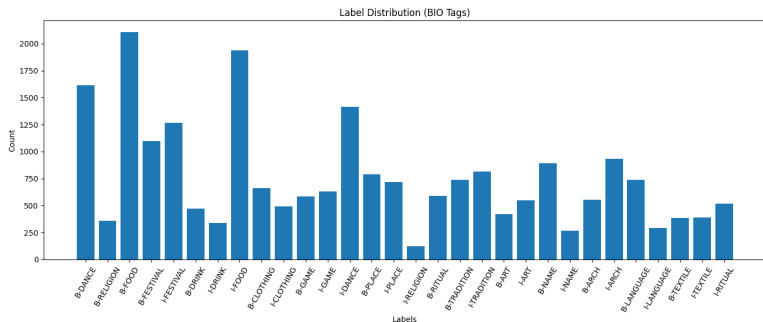
Gondi → **LANGUAGE**

Detailed BIO Example

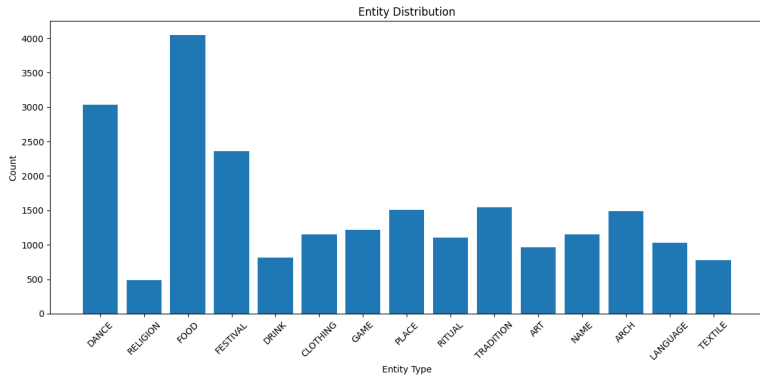
Tokens: [gondi, is, part, of, indian, tradition]

Tags: [B-LANGUAGE, O, O, O, O, O]

Label Distribution



Entity Distribution



- Model: **BERT (bert-base-uncased)**
- Task: Token Classification (NER)
- Fine-tuned on cultural dataset
- Optimizer: AdamW
- Epochs: 5
- Learning rate: $2e-5$

Results

Performance

F1 Score: **0.8649**

Loss: 0.191

- High F1-score indicates accurate predictions
- Performs well on unseen test data
- Handles multiple entity categories effectively

Analysis

- Data augmentation improved performance
- Balanced dataset reduced bias
- BIO tagging captures multi-word entities
- Context-aware learning via BERT improved accuracy

Limitation

Synthetic sentences may not fully represent real-world data

Team Contributions

- **Dinesh**: Research and literature study
- **Ajay**: Dataset generation and preprocessing
- **Dhanush**: Model implementation and training

Conclusion

Summary

- Built a Cultural NER system
- Achieved F1-score of 0.86
- Demonstrated effectiveness of data augmentation

Thank You